5-Bar SCARA v2 Onboarding Package

M1

Encoder + Sensor Fusion Edition | Engineering Roadmap

MECHANICAL LEAD

1 | ROLE & OBJECTIVE

As **Mechanical Lead (M1)**, you are the geometric and physical foundation of the entire project. Your mission is two-fold: **extract precise kinematic parameters** (L1–L4, d) to ±0.1 mm, and **physically mount and validate the rotary encoders** on both NEMA 17 motor shafts. Every downstream firmware component — IK solver, closed-loop correction, collision detection — depends on the accuracy of your measurements. You own Phase 1 outright and provide hardware support for Phase 2B.

Primary Mission: Deliver a fully-populated **config.h** with all geometric and encoder constants, verified by oscilloscope, and committed to Git before M2 can begin Phase 2B.

2 | HARDWARE & COMPONENTS REQUIRED

Component	Specification / Notes	Quantity
NEMA 17 Stepper Motors	Already installed on SCARA frame	2
Rotary Encoder (Magnetic)	AS5600 I2C OR LPD3806-600BM-G5-24C quadrature, 600 PPR	2
Encoder Mounting Brackets	Custom-printed or machined; must be rigid (zero flex)	2
Digital Calipers	Resolution ±0.01 mm; calibrated	1
Dial Indicator + Magnetic Stand	For runout check — must verify < 0.1 mm eccentricity	1
Oscilloscope / Logic Analyzer	To verify clean A/B quadrature square waves at full speed	1
Decoupling Capacitors	100 nF ceramic; 1 per encoder signal line (4 total)	4+
Shielded Signal Cable	For A/B encoder channels; route away from motor power leads	As needed
Alignment Tape / Marker	Physical home-position marking on workspace	1
Screws, Standoffs, Epoxy	For rigid bracket + motor housing attachment	Assorted

3 | SOFTWARE & PLATFORMS REQUIRED

Tool / Library	Purpose	Installation
Arduino IDE 2.x or PlatformIO	Edit and verify config.h constants	arduino.cc / platformio.org
ESP32 Board Package (Espressif)	For ESP32 pin definitions referenced in config.h	Board Manager URL
Git (CLI or GitHub Desktop)	Version control — commit v0.0-baseline and config.h	git-scm.com
VSCode + C/C++ Extension	Syntax highlighting for config.h editing	code.visualstudio.com
Logic Analyzer Software	Pulseview (open source) to decode quadrature signals	sigrok.org
Spreadsheet (Excel / Sheets)	Record 5x measurement sets and compute means	Office / Google

4 | AI INTEGRATION STRATEGY

Use AI to validate your measurements, automate config.h generation, and troubleshoot encoder wiring. Below are specific, copy-paste-ready prompt strategies:

Prompt 1 — Generate config.h from your measurements:

"I measured the 5-bar SCARA linkage parameters: L1=118.4mm, L2=120.1mm, L3=117.9mm, L4=119.6mm, d=95.0mm. The encoder is LPD3806 with 600 PPR using quadrature x4 decoding. Generate a complete config.h with all constexpr values, GPIO pin assignments for ESP32 interrupt-capable pins, and inline comments explaining each constant. Format it as Arduino C++."

Prompt 2 — Diagnose encoder signal quality:

"I see my quadrature encoder A/B signals on oscilloscope. The A channel has clean edges but B channel has a 50ns ringing glitch after each edge. My wiring runs parallel to the stepper motor coil wires for ~15cm. What are the likely causes and fixes? Explain EMI coupling and decoupling capacitor placement."

Prompt 3 — Validate bracket design:

"I need to mount a rotary encoder to a NEMA 17 motor shaft. The encoder body must not flex. List the mechanical design requirements for the bracket, runout constraints, and how to verify concentricity with a dial indicator. Provide a step-by-step procedure."

5 | BASIC ARCHITECTURE & CONNECTIONS

Your Deliverable	Connects To	Why It Matters
config.h (L1–L4, d)	M2 — IK Solver (Phase 2)	IK math uses these exact values; 1mm error = positioning fault
Encoder PPR, GPIO pins	M2 — Encoder ISR (Phase 2B)	ISR must know which GPIO and CPR to decode counts correctly
Physical home marker	M6 — Baseline Audit (Pre-Phase)	M6 needs reference position before Git commit
Encoder wire routing	M5 — Reviewer (Phase 2B)	M5 checks EMI isolation and signal integrity
Oscilloscope verification	M5 — TEST 1-D sign-off	Gate requirement; no clean signal = Phase 1 blocked

Your Deliverable	Connects To	Why It Matters
Rigid bracket + runout < 0.1mm	M2 — Closed-loop accuracy	Mechanical slop propagates into position error directly

Data Flow: M6 baseline commit → **M1 physical measurements + encoder mount** → M2 IK firmware → M2 encoder ISR → M4 integration → M5/M6 validation

6 | TIMELINE & MILESTONES

[DAY 1]

- Attend kickoff with M6 — confirm Git repo access and review v0.0-baseline tag
- Gather all measurement tools (calipers, dial indicator, oscilloscope)
- Print or acquire encoder mounting brackets; order encoders if not in stock
- Review config.h skeleton from M6's Pre-Phase deliverable

[DAYS 2–3 (Phase 1 Core)]

- Measure L1, L2, L3, L4 five times each — record in spreadsheet, compute means
- Measure d (motor pivot separation) five times — record mean
- Measure joint backlash in ±X and ±Y with dial indicator
- Mark and photograph world frame origin on workspace surface

[DAYS 3–4 (Encoder Mounting — NEW v2)]

- Mount encoder body to motor housing using rigid bracket (zero flex)
- Attach encoder disk to motor shaft — verify concentricity with dial indicator (runout < 0.1mm)
- Route A/B cables AWAY from motor power wires; add 100nF decoupling caps at ESP32 pins
- Connect encoder to ESP32 on interrupt-capable pins (ENC1_A=34, ENC1_B=35, ENC2_A=36, ENC2_B=39)

[DAY 5 (Verification Gate)]

- Oscilloscope check: clean square waves on A and B at full motor speed — no glitches
- Populate config.h: L1–L4, d, ENCODER_PPR=600, ENC_CPR=2400, GPIO pin assignments
- Commit config.h to Git — coordinate with M6 for tag/branch naming
- M5 reviews: TEST 1-A through 1-E — obtain sign-off

[PHASE 2B HW SUPPORT]

- Support M2 during encoder homing: zero both encoder counts at physical home position
- Verify 3 known angles physically — confirm encoder within ±2 counts of expected
- If stall detection triggers incorrectly: check for mechanical binding, tighten bracket
- Sign off on Phase 2B HW portion with M5 and M2

7 | ESCALATION MATRIX

Issue / Blocker	First Contact	Second Contact	Notes
Measurement ambiguity / linkage geometry unclear	M6 (Systems Lead)	M2 (uses values in IK)	Remeasure and document disagreement
Encoder signal glitches persist after decoupling caps	M2 (Firmware)	M5 (Test Lead)	Try ferrite bead on signal lines
Encoder bracket flexes — runout > 0.1mm	Self-resolve: redesign bracket	M6 for fabrication resources	Do NOT proceed to firmware until resolved

Issue / Blocker	First Contact	Second Contact	Notes
GPIO pin conflict with M2's firmware	M2 (pin ownership)	M6 (config arbitration)	Update config.h together
Git merge conflict on config.h	M6 (Git custodian)	M4 (Integration)	M6 resolves baseline conflicts
Phase gate blocked — M5 not available for review	M5 directly	M6 for escalation	Gate CANNOT be bypassed

You are the physical foundation of this system. Every millimeter of precision you deliver multiplies through the entire kinematic chain. Measure twice, commit once. You've got this. ■